

Guide to Typst

May 25, 2023 - v0.1 - for typist v0.4.0

tschinz

whynotlogic@gmail.com

HEI-Vs

Contents

1 Introduction	4
2 Installation	5
2.1 With cargo	5
2.2 MacOS	5
2.3 Linux	5
3 Formatting	6
3.1 Markup	6
3.2 Page Formatting	6
3.3 Space	6
3.4 Text Formatting	6
4 Elements	10
4.1 Headings	10
4.2 Lists	10
4.3 Custom Lists	11
4.4 Images	11
4.4.1 Alignment	11
4.4.2 Caption	11
4.4.3 Cluster	12
4.5 Tables	13
4.6 Icon Boxes	15
4.7 Color Boxes	16
5 References	18
5.1 Links	18
5.2 Crossreferences	18
5.3 External References	18
5.4 Glossary	18
5.5 Acronym	19
6 Code	20
7 Math Equations	21
7.1 Align	21
7.2 Symbols	21
7.2.1 Accents	21
7.2.2 Equals & Operators	22

7.2.3 Scripts	22
7.2.4 Special Elements	22
7.2.5 Alphabeth	23
7.2.6 Logical	24
7.2.7 Operators	24
7.2.8 Arrows	24
7.2.9 Angles	25
7.2.10 Cool Symbols	26
7.2.11 Style	26
8 Emoji Symbols	27
Bibliography	28

Figures

Figure 1: ZNotes Icon	12
Figure 2: Multiple images one caption	12
Figure 3: Multiple images one caption	12
Figure 4: Caption left image	13
Figure 5: Caption right image	13
Figure 6: Caption topleft image	13
Figure 7: Caption topright image	13
Figure 8: Caption bottomleft image	13
Figure 9: Caption bottomright image	13
Figure 10: Some proof	21

Tables

Table 1: Table caption	14
Table 2: Links	18

Listings

Listing 1: Label inserts	18
Listing 2: Rust Code	20

Equation

Equation 1 21

Equation 2 21

Equation 3 21

Equation 4 21

Equation 5 22

Equation 6 22

Equation 7 22

Equation 8 22

Equation 9 23

Equation 10 23

Equation 11 23

Equation 12 23

Equation 13 23

Equation 14 23

Equation 15 23

Equation 16 23

Equation 17 23

Equation 18 23

Equation 19 23

Equation 20 23

Equation 21 23

Equation 22 23

Equation 23 23

1 | Introduction

The goal of this document is to have the most common used elements for the markup language typst readily available. A detailed documentation can be found on their website: <https://typst.app/docs> It is to note that these are **my** most common used elements. For some elements custom templates are needed:

- [tablex](#)
- [myref](#)
- all files in the 00-templates/ folder such as
 - boxes.typ
 - constants.typ
 - helpers.typ
 - items.typ
 - metadata.typ
 - template-*

2 | Installation

2.1 With cargo

If you use already the rust programming language then you can use rust to install the latest toolchain.

```
# install rust and cargo
curl https://sh.rustup.rs -sSf | sh

# install typst
cargo install --git https://github.com/typst/typst
```

2.2 MacOS

On MacOS you can sue homebrew

```
# install homebrew
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"

# install typst
brew install typst
```

2.3 Linux

In Linux you can use the commonly available package manager

```
brew install typst
pacman -S typst
xbps-install typst
sudo apt-get install typst
```

3 | Formatting

3.1 Markup

Name	Example	Raw
Singleline Comment		//
Multiline Comment		/* */
Paragraph break		blankline
Line break		\
bold	bold	*bold*
italic	<i>italic</i>	_italic_
monospaced	monospaced	`monospaced`
math	$x = 1$	$x=1$
lowercase	lower	#lower("LoWeR")
uppercase	UPPER	#upper("UpPeR")
smallcaps	SMALLCAPS	#SmallCaps("smallcaps")
smartquote	“test”	#smartquote()test#smartquote()
overline	<u>overline</u>	#overline("overline")
underline	<u>underline</u>	#underline("underline")
strike	strike	#strike("strike")
sub	Text _{sub}	Text#sub("sub")
super	Text ^{super}	Text#super("super")
Label		<label>
Reference		@label

3.2 Page Formatting

```
#pagebreak() // pagebreak
#parbreak()  // parbreak
\           // linebreak
```

3.3 Space

A

B

A #h(5cm) B,

C

D

C #v(0.2cm) D

3.4 Text Formatting

For the custom textsizes and colors you need to import:

```
#import "../01-tail/constants.typ": *
```

Name	Example	Raw
Sizes	8pt text tiny text	text(8pt, "8pt text") text(tiny "tiny text")
	9pt text smaller text	text(9pt, "9pt text") text(smaller "smaller text")
	10pt text small text	text(10pt, "10pt text") text(small "small text")
	11pt text normal text	text(11pt, "11pt text") text(normal "normal text")
	14pt text large text	text(14pt, "14pt text") text(large "large text")
	16pt text larger text	text(16pt, "16pt text") text(larger "larger text")
	24pt text huge text	text(24pt, "24pt text") text(huge "huge text")
Types	36pt text huger text	text(36pt, "36pt text") text(huger "huger text")
	Fira Sans	text(font:"Fira Sans", "Fira Sans")
	Fira Mono	text(font:"Fira Mono", "Fira Mono")
	Source Sans Pro	text(font:"Source Sans Pro", "Source Sans Pro")
	New Computer Modern	text(font:"New Computer Modern", "New Computer Modern")
	New Computer Modern Sans	text(font:"New Computer Modern Sans", "New Computer Modern Sans")

Alignment	start	align(start){start}
	end	align(end){end}
	left	align(left){left}
	center	align(center){center}
	right	align(right){right}
	top	align(top){top}
	horizon	align(horizon){horizon}
	bottom	align(bottom){bottom}
	center + horizon	align(center + horizon){center + horizon}

Colors	black	#text(fill:black)[black]
	red	#text(fill:red)[red]
	green	#text(fill:green)[green]
	blue	#text(fill:blue)[blue]
	purple	#text(fill:purple)[purple]
	gray-80	#text(fill:gray-80)[gray-80]
	gray-70	#text(fill:gray-70)[gray-70]
	gray-60	#text(fill:gray-60)[gray-60]
	gray-50	#text(fill:gray-50)[gray-50]
	gray-40	#text(fill:gray-40)[gray-40]
	gray-30	#text(fill:gray-30)[gray-30]
	gray-20	#text(fill:gray-20)[gray-20]
	gray-10	#text(fill:gray-10)[gray-10]
	hei-orange	#text(fill:hei-orange)[hei-orange]
	hei-blue	#text(fill:hei-blue)[hei-blue]
	hei-pink	#text(fill:hei-pink)[hei-pink]
	hei-yellow	#text(fill:hei-yellow)[hei-yellow]
	hei-green	#text(fill:hei-green)[hei-green]
	spl-green	#text(fill:spl-green)[spl-green]
	spl-blue	#text(fill:spl-blue)[spl-blue]
	spl-pink	#text(fill:spl-pink)[spl-green]
	color-info	#text(fill:color-info)[color-info]
	color-idea	#text(fill:color-idea)[color-idea]
	color-warning	#text(fill:color-warning)[color-warning]
	color-important	#text(fill:color-important)[color-important]
	color-fire	#text(fill:color-fire)[color-fire]
	color-rocket	#text(fill:color-rocket)[color-rocket]
	color-todo	#text(fill:color-todo)[color-todo]
	code-bg	#text(fill:code-bg)[code-bg]
	code-border	#text(fill:code-border)[code-border]

4 | Elements

4.1 Headings

```
= Heading 1
== Heading 1.1
=== Heading 1.1.1
==== Heading 1.1.1.1
...
```

4.2 Lists

- First
- Second
- Third

```
- First
- Second
- Third
```

- First
 - Second
 - Third

```
- First
- Second
- Third
```

- First
- Second
- Third

```
- First
- Second
- Third
```

- First
- Second
- Third

```
list(
  [First],
  [Second],
  [Third],
)
```

1. First
 1. Second
2. Third

```
+ First
+ Second
+ Third
Text
4. Fourth
+ Fifth
```

Text

4. Fourth
5. Fifth

1. First
 - a) Second
2. Third

```
+ First
  #set enum(numbering: "a")
+ Second
+ Third
Text
4. Fourth
+ Fifth
```

Text

4. Fourth
5. Fifth

4.3 Custom Lists

```
#import "../01-head/items.typ": *
```

- ☑ item-list
- 🔖 item-checkbadge
- ⊙ item-checkcircle
- ☑ item-checksquare
- ✓ item-check
- 📁 item-file
- 📁 item-folder
- ⊗ item-xcircle
- ⊗ item-xsquare
- × item-x

```
#item-list(content:"item-list")
#item-checkbadge(content:"item-checkbadge")
#item-checkcircle(content:"item-checkcircle")
#item-checksquare(content:"item-checksquare")
#item-check(content:"item-check")
#item-file(content:"item-file")
#item-folder(content:"item-folder")
#item-xcircle(content:"item-xcircle")
#item-xsquare(content:"item-xsquare")
#item-x(content:"item-x")
```

4.4 Images

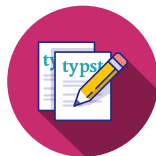
4.4.1 Alignment

left



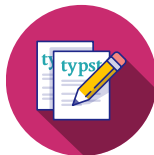
```
#image("../04-resources/icon.svg",
width: 2cm)
```

center



```
#align(center,
image("../04-resources/icon.svg",
width: 2cm)
)
```

right



```
#align(right,
image("../04-resources/icon.svg",
width: 2cm)
)
```

4.4.2 Caption

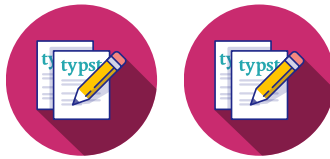


Figure 1: ZNotes Icon

```
#figure(
  image("../04-resources/icon.svg",
    width: 2cm),
  caption: [ZNotes Icon]
) <fig-icon>
```

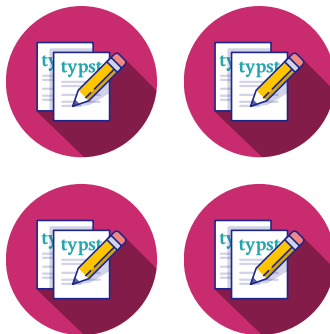
4.4.3 Cluster

Two images one caption

Figure 2: Multiple images **one** caption

```
#figure(
  tablex(
    columns: 2,
    stroke: none,
    align: center + horizon,
    image(icon, width: 2cm), image(icon, width: 2cm)
  ),
  caption: [Multiple images *one* caption]
)
```

Four images one caption

Figure 3: Multiple images **one** caption

```
#figure(
  tablex(
    columns: 2,
    stroke: none,
    align: center + horizon,
    image(icon, width: 2cm), image(icon, width: 2cm),
    image(icon, width: 2cm), image(icon, width: 2cm),
  ),
  caption: [Multiple images *one* caption]
)
```

```
caption: [Multiple images *one* caption]
)
```

Two images two caption

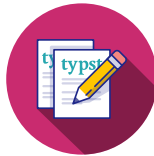


Figure 4: Caption left image



Figure 5: Caption right image

```
#align(center,
  tablex(
    columns: 2,
    stroke: none,
    align: center + horizon,
    figure(image(icon, width: 2cm), caption: [Caption left image]), figure(image(icon,
width: 2cm), caption: [Caption right image]),
  ))
```

Four images four caption

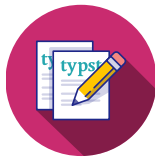


Figure 6: Caption topleft image

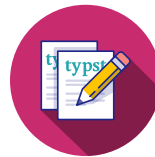


Figure 7: Caption topright image

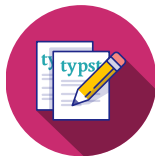


Figure 8: Caption bottomleft image



Figure 9: Caption bottomright image

```
#align(center,
  tablex(
    columns: 2,
    stroke: none,
    align: center + horizon,
    figure(image(icon, width: 2cm), caption: [Caption topleft image]),
    figure(image(icon, width: 2cm), caption: [Caption topright image]),
    figure(image(icon, width: 2cm), caption: [Caption bottomleft image]),
    figure(image(icon, width: 2cm), caption: [Caption bottomright image]),
  ))
```

4.5 Tables

For all `#tablex` command the appropriate module needs to be imported

```
#import "../01-head/tablex.typ": *
```

Tables with and without caption

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

Table 1: Table caption

```
tablex(  
  columns: 3,  
  align: center + horizon,  
  [  
    , [*Col1*] , [*Col2*],  
  [*Row1*], "cell-0-0", "cell-1-0",  
  [*Row2*], "cell-0-1", "cell-1-1",  
  ]  
)
```

```
figure(  
  tablex(  
    columns: 3,  
    align: center + horizon,  
    [  
      , [*Col1*] , [*Col2*],  
    [*Row1*], "cell-0-0", "cell-1-0",  
    [*Row2*], "cell-0-1", "cell-1-1",  
    ],  
    kind: table,  
    caption: [Table Caption]  
  )  
)
```

Tables with cell spans

	Col1	Col2
Row1	cell-0	cell-1-0
Row2		cell-1-1

	Col1	Col2
Row1	cell-0	
Row2	cell-0-1	cell-1-1

```
tablex(  
  columns: 3,  
  align: center + horizon,  
  [  
    , [*Col1*] , [*Col2*],  
  [*Row1*], rowspanx(2)[cell-0],  
  "cell-1-0",  
  [*Row2*], "cell-1-1",  
  ]  
)
```

```
tablex(  
  columns: 3,  
  align: center + horizon,  
  [  
    , [*Col1*] , [*Col2*],  
  [*Row1*], colspanx(2)[cell-0],  
  [*Row2*], "cell-0-1", "cell-1-1",  
  ]  
)
```

Table Design

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

```
tablex(  
  columns: 3,  
  align: center + horizon,  
  [  
    , [*Col1*] , [*Col2*],  
  [*Row1*], "cell-0-0", "cell-1-0",  
  [*Row2*], "cell-0-1", "cell-1-1",  
  ]  
)
```

```
tablex(  
  columns: 3,  
  align: center + horizon,  
  [  
    , [*Col1*] , [*Col2*],  
  [*Row1*], "cell-0-0", "cell-1-0",  
  [*Row2*], "cell-0-1", "cell-1-1",  
  ]  
)
```

```

auto-vlines: false,
align: center + horizon,
[[] , [*Col1*] , [*Col2*],
[*Row1*], "cell-0-0", "cell-1-0",
[*Row2*], "cell-0-1", "cell-1-1",
)

```

```

auto-hlines: false,
align: center + horizon,
[[] , [*Col1*] , [*Col2*],
[*Row1*], "cell-0-0", "cell-1-0",
[*Row2*], "cell-0-1", "cell-1-1",
)

```

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

```

#tablex(
  columns: 3,
  auto-lines: false,
  align: center + horizon,
  (), vlinex(stroke: blue) , vlinex(), (),
  [[] , [*Col1*] , [*Col2*], hlinex(stroke: red),
  [*Row1*], "cell-0-0", "cell-1-0", hlinex(),
  [*Row2*], "cell-0-1", "cell-1-1",
)

```

4.6 Icon Boxes

```
#import "../01-head/boxes.typ": *
```



```
#infobox()["infobox"]
```



```
#ideabox()["ideabox"]
```



```
#warningbox()["warningbox"]
```



```
#importantbox()["importantbox"]
```



```
#firebox()["firebox"]
```



```
#rocketbox()["rocketbox"]
```



```
#todobox()["todobox"]
```



```
#box(icon:"../04-resources/placeholder.svg", linecolor: hei-  
pink)["todobox"]
```

4.7 Color Boxes

```
#import "../01-head/boxes.typ": *
```

Exercise

Some text

```
#colorbox( title: "Exercise", color:hei-blue)[Some text]
```

Attention

Some text

```
#colorbox( title: "Attention", color:hei-pink)[Some text]
```

Consider

Some text

```
#colorbox( title: "Consider", color:hei-green)[Some text]
```

Information

Some text

```
#colorbox( title: "Information", color:hei-orange)[Some text]
```

5 | References

5.1 Links

Example	Raw
https://example.com	<code>https://example.com</code>
https://example.com	<code>#link("https://example.com")</code>
See example.com	<code>#link("https://example.com")[See example.com]</code>
whynotlogic@gmail.com	<code>#link("mailto:whynotlogic@gmail.com")[whynotlogic\@gmail.com]</code>
	<code>#link("https://tschinz.github.io/znotes")[#image(icon, width:0.5cm)]</code>

Table 2: Links

5.2 Crossreferences

In the document the following references were added.

```
= References <sec-ref>
== Links <sec-links>
#figure(image("../04-resources/icon.svg", width: 2cm)) <fig-icon>
#figure(tablex(...), kind:table) <tab-links>
#figure(align(left, raw(...)) <code-ref>
$ sum_(k=1)^n k = (n(n+1)) / 2 $ <math-eq1> #ref(<math-eq1>)
```

Listing 1: Label inserts

They can be references as follows:

Type	Example	Raw
Section	Section 5	<code>@sec-ref</code>
Subsection	Section 5.1	<code>@sec-links</code>
Table	Table 2	<code>@tab-links</code>
Code	Listing 1	<code>@code-ref</code>

5.3 External References

Example

[1]

[1]

Raw

```
#cite("stateoftheArt")
```

```
@stateoftheArt
```

5.4 Glossary

The glossary entries need to be defined in `03-tail/glossary.typ`. For the glossary functions the “import” of `01-head/helpers.typ` is needed.

```
#import "../01-head/helpers.typ": *
#import "../03-tail/glossary.typ": *
```

Example**Raw**

Scrum

```
#gls-scrumm.name
```

Scrum is an agile process framework for managing complex knowledge work, with an initial emphasis on software development, although it has been used in other fields and is slowly starting to be explored for other complex work, research and advanced technologies.

```
#gls-scrumm.description
```

5.5 Acronym

The acronym entries need to be defined in 03-tail/glossary.typ. For the acronym functions the “import” of 01-head/helpers.typ is needed.

```
#import "../01-head/helpers.typ": *
#import "../03-tail/glossary.typ": *
```

Example**Raw**

AR

```
#acr-ar.abr
```

AR

```
#acrshort(acr-ar.abbr)
```

Augmented Reality

```
#acr-ar.long
```

Augmented Reality)

```
#acrlong(acr-ar)
```

Augmented Reality (AR)

```
#acr-ar.long (#acr-ar.abbr)
```

Augmented Reality (AR)

```
#acrfull(acr-ar)
```

6 | Code

inline monospaced string

```
`inline monospaced string`
```

```
fn main() {println!("Hello world!")}
```

```
raw(lang:"rust",  
  "fn main() {println!(\"Hello world!  
  \")\"  
  })
```

```
fn main() {  
  println!("Hello world!")  
}
```

```
``rust  
fn main() {  
  println!("Hello world!")  
}  
``
```

```
fn main() {  
  println!("Hello world!")  
}
```

```
#figure(  
  align(left,  
    ``rust  
    fn main() {  
      println!("Hello world!")  
    }  
    ``  
  ),  
  caption: [Rust Code],  
)
```

Listing 2: Rust Code

7 | Math Equations

Inline math

Let a and b , and c be the side of a right-angled triangle.

Let a and b , and c be the side of a right-angled triangle.

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$\sum_{k=1}^n k = (n(n+1)) / 2$,

Fullline math

$$a^2 + b^2 = c^2 \tag{1}$$

$a^2 + b^2 = c^2$ `<math-eq1>`

Math with caption

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \tag{2}$$

Figure 10: Some proof

```
#figure(
  $ sum_(k=1)^n k = (n(n+1)) / 2 $,
  caption: [Some proof]
)
```

7.1 Align

Formula

$$\begin{aligned} a_1 &= b_1 + c_1 = z_1 \\ a_2 &= b_2 + c_2 - d_2 + e_2 = z_1 \end{aligned} \tag{3}$$

Raw

```
$
a_1 = b_1 + c_1 = z_1 \ \
a_2 = b_2 + c_2 - d_2 + e_2 = z_1
$
```

$$\begin{aligned} a_1 &= b_1 + c_1 &&= z_1 \\ a_2 &= b_2 + c_2 - d_2 + e_2 = z_1 \end{aligned} \tag{4}$$

```
$
a_1 &= b_1 + c_1 &= z_1 \ \
a_2 &= b_2 + c_2 - d_2 + e_2 &= z_1
$
```

7.2 Symbols

This is an incomplete list for all symbols goto [here](#)

Outside of the $\$$ math environment the symbols can be accessed with `.`

7.2.1 Accents

Symbol Raw $\grave{x}$ `$grave(x)$` $\tilde{x}$ `$tilde(x)$` $\ddot{x}$ `$dot.double(x)$` $\ddot{x}$ `$diaer(x)$` $\check{x}$ `$caron(x)$` $\cancel{x}$ `$cancel(x)$` $\overline{xyz}$ `$overline(xyz)$` $\underbrace{xyz}$ `$underbracket(xyz)$`**Symbol Raw** $\acute{x}$ `$acute(x)$` $\breve{x}$ `$breve(x)$` triple dot `$dot.triple(x)$` $\circ$ `$circle(x)$` $\vec{x}$ `$arrow(x)$` $\bar{x}$ `$macron(x)$` $\underbrace{xyz}$ `$underbrace(xyz)$` $\overbrace{xyz}$ `$overbracket(xyz)$`**Symbol Raw** $\hat{x}$ `$hat(x)$` $\dot{x}$ `$dot(x)$` quad dot `$dot.quad(x)$` acute double `$acute.double(x)$` $\vec{x}$ `$arrow.l(x)$` $\overline{xyz}$ `$overline(xyz)$` $\overbrace{xyz}$ `$overbrace(xyz)$` $\overbrace{xyz}$ `$overbracket(xyz)$`**7.2.2 Equals & Operators****Symbol Raw** $=$ `$=$` $\neq$ `$!=$` $\simeq$ `$tilde.eq$` $\geq$ `$gt.eq$` $\nless$ `$lt.eq.not$` $\not\approx$ `$approx.not$` $=:$ `$eq.colon$` $+$ `$plus$` $\oplus$ `$plus.circle$` $\mp$ `$minus.plus$`**Symbol Raw** $=$ `$eq$` $\equiv$ `$ident$` $\not\sim$ `$tilde.eq.not$` $\nless$ `$gt.eq.not$` $\approx$ `$approx$` $:$ `$colon$` $::=$ `$colon.double.eq$` $+$ `$plus.small$` $-$ `$-$` $\ominus$ `$minus.circle$`**Symbol Raw** $\neq$ `$eq.not$` $\equiv$ `$ident.strict$` $=$ `$eq.small$` $\leq$ `$lt.eq$` $\approx$ `$approx.eq$` $:=$ `$colon.eq$` $+$ `$+$` $\pm$ `$plus.minus$` $-$ `$minus$`**7.2.3 Scripts****Symbol Raw** x_1 `$x_1$` x_1 `$x_1$` x_1^2 `$x_1^2$` $\overset{2}{x}_1$ `$x_1^2$`**Symbol Raw** x_{12} `$x_(12)$` x_{12} `$x_(12)$` x_{12}^{34} `$x_(12)^(34)$` x_{12}^{34} `$x_(12)^(34)$`**Symbol Raw** x_1 `$scripts(x)_1$` x_1 `$scripts(x)_1$` x_1^2 `$scripts(x)_1^2$` x_1^2 `$scripts(x)_1^2$`**7.2.4 Special Elements****Symbol****Raw** $\binom{n}{k}$ (5) `$ binom(n, k) $` $\left[\frac{1}{2}\right]$ (7) `$ round(1, 2) $`**Symbol****Raw** $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ (6) `$ vec(1, 2, delim: "[") $` $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ (8) `$ mat(1,2; 3,4) $`

$$\begin{pmatrix} 1 & 2 & \dots & 10 \\ 2 & 2 & \dots & 10 \\ \vdots & \vdots & \ddots & \vdots \\ 10 & 10 & \dots & 10 \end{pmatrix} (9)$$

```
$ mat(
  1, 2, ..., 10;
  2, 2, ..., 10;
          dots.v,
dots.v, dots.down,
dots.v;
  10, 10, ...,
  10;
) $
```

$$\sum a_k \quad (10) \quad \$ \text{sum } a_k \$$$

$$\sum_{k=0}^n a_k \quad (11) \quad \$ \text{sum_}(k=0)^n a_k \$$$

$$\sum_{k=0}^n a_k \quad (12) \quad \$ \text{scripts}(\text{sum})_ (k=0)^n a_k \$$$

$$\sqrt[3]{x} \quad (13) \quad \$ \text{root}(3, x) \$$$

$$f(x, y) := \begin{cases} 1 & \text{if } \frac{x \cdot y}{2} \leq 0 \\ 2 & \text{if } x \text{ is even} \\ 3 & \text{if } x \in \mathbb{N} \\ 4 & \text{else} \end{cases} \quad (14)$$

```
$ f(x, y) := cases(
  1 "if" (x dot y)/2 <= 0,
  2 "if" x "is even",
  3 "if" x in NN,
  4 "else",
) $
```

$$\frac{1}{2} \quad (15) \quad \$ 1/2 \$$$

$$\frac{1}{2} \quad (16) \quad \$ \text{frac}(1,2) \$$$

$$\frac{x+1}{x+2} \quad (17) \quad \$ (x+1)/(x+2) \$$$

$$\frac{(x+1)}{(x+2)} \quad (18) \quad \$ ((x+1))/((x+2)) \$$$

$$\prod \quad (19) \quad \$ \text{product} \$$$

$$n! = \prod_{k=1}^n k \quad (20) \quad \$ \text{product_}(k=1)^n k \$$$

$$n! = \prod_{k=1}^n k \quad (21) \quad \$ n! = \text{scripts}(\text{product})_ (k=1)^n k \$$$

$$\int \quad (22) \quad \$ \text{integral} \$$$

$$\int_a^b f(x) \quad (23) \quad \$ \text{integral} \$$$

7.2.5 Alphabeth

Symbol

$\alpha\beta\gamma\delta\epsilon\zeta\eta\theta\iota\kappa\lambda\mu\nu\xi\omicron\rho\sigma\tau\upsilon\varphi\chi\psi\omega$

ΑΒΓΔΕΖΗΘΙΚΑΜΝΞΟΠΡΣΤΥΦΧΨΩ

ABCDEFGHIJKLMNOPQRSTUVWXYZ

Raw

α β γ δ ϵ ζ η θ ι κ λ μ ν ξ $\omicron$ π ρ σ τ υ φ χ ψ ω

α β γ δ ϵ ζ η θ ι κ λ μ ν ξ $\omicron$ π ρ σ τ υ φ χ ψ ω

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

7.2.6 Logical

Symbol	Raw	Symbol	Raw	Symbol	Raw
$\wedge$	<code>\$and\$</code>	$\bigwedge$	<code>\$and.big\$</code>	$\&$	<code>\$amp\$</code>
$\vee$	<code>\$or\$</code>	$ $	<code>\$bar.v\$</code>	$*$	<code>\$ast.op\$</code>
$*$	<code>\$ast.basic\$</code>	$*$	<code>\$ast.low\$</code>	$\oplus$	<code>\$plus.circle\$</code>
$\oplus$	<code>\$plus.circle.big\$</code>				

7.2.7 Operators

Symbol	Raw	Symbol	Raw	Symbol	Raw
$\sin x$	<code>\$sin x\$</code>	$\cos x$	<code>\$cos x\$</code>	$\tan x$	<code>\$tan x\$</code>
$\arcsin x$	<code>\$arcsin x\$</code>	$\arccos x$	<code>\$arccos x\$</code>	$\arctan x$	<code>\$arctan x\$</code>
$\sinh x$	<code>\$sinh x\$</code>	$\cosh x$	<code>\$cosh x\$</code>	$\tanh x$	<code>\$tanh x\$</code>
$\arg x$	<code>\$arg x\$</code>	$\csc x$	<code>\$csc x\$</code>	$\deg x$	<code>\$deg x\$</code>
$\det x$	<code>\$det x\$</code>	$\dim x$	<code>\$dim x\$</code>	$\exp x$	<code>\$exp x\$</code>
$\bmod x$	<code>\$mod x\$</code>	$\inf x$	<code>\$inf x\$</code>	$\log x$	<code>\$log x\$</code>
$\lim x$	<code>\$lim x\$</code>	$\liminf x$	<code>\$liminf x\$</code>	$\limsup x$	<code>\$limsup x\$</code>
$\min x$	<code>\$min x\$</code>	$\max x$	<code>\$max x\$</code>	$\sup x$	<code>\$sup x\$</code>

7.2.8 Arrows

SymRaw	SymRaw	SymRaw	
	Arrows right		
$\rightarrow$	<code>\$arrow\$</code>	$\mapsto$	<code>\$arrow.bar\$</code>
$\mapsto$	<code>\$arrow.bar.long\$</code>	$\Rightarrow$	<code>\$arrow.double.long\$</code>
$\mapsto$	<code>\$arrow.double.bar\$</code>	$\Rightarrow$	<code>\$arrow.double.bar.long\$</code>
$\Rightarrow$	<code>\$arrow.stroked\$</code>	$\Rightarrow$	<code>\$arrow.quad\$</code>
$\hookrightarrow$	<code>\$arrow.curve\$</code>	$\dashrightarrow$	<code>\$arrow.dashed\$</code>
	$\rightsquigarrow$	$\looparrowright$	<code>\$arrow.loop\$</code>
	Arrows left		
$\leftarrow$	<code>\$arrow.l\$</code>	$\longleftarrow$	<code>\$arrow.l.long\$</code>
$\longleftarrow$	<code>\$arrow.l.bar.long\$</code>	$\longleftarrow$	<code>\$arrow.l.bar\$</code>
$\longleftarrow$	<code>\$arrow.l.double\$</code>	$\longleftarrow$	<code>\$arrow.l.double.long\$</code>
$\longleftarrow$	<code>\$arrow.l.double.bar\$</code>	$\longleftarrow$	<code>\$arrow.l.double.bar.long\$</code>
$\longleftarrow$	<code>\$arrow.l.quad\$</code>	$\longleftarrow$	<code>\$arrow.l.quad\$</code>
$\longleftarrow$	<code>\$arrow.l.stroked\$</code>	$\longleftarrow$	<code>\$arrow.l.dashed\$</code>
$\longleftarrow$	<code>\$arrow.l.curve\$</code>	$\longleftarrow$	<code>\$arrow.l.loop\$</code>
	$\rightsquigarrow$		
	Double Arrows Left Right		
$\leftrightarrow$	<code>\$arrow.l.r\$</code>	$\nleftrightarrow$	<code>\$arrow.l.r.not\$</code>
$\longleftrightarrow$	<code>\$arrow.l.r.double\$</code>	$\longleftrightarrow$	<code>\$arrow.l.r.long\$</code>
$\longleftrightarrow$	<code>\$arrow.l.r.double.long\$</code>	$\longleftrightarrow$	<code>\$arrow.l.r.double.not\$</code>
$\longleftrightarrow$	<code>\$arrow.l.r.stroked\$</code>	$\longleftrightarrow$	<code>\$arrow.l.r.filled\$</code>
		$\longleftrightarrow$	<code>\$arrow.l.r.wave\$</code>

Arrows Top `$arrow.t$` `$arrow.t.triple$` `$arrow.t.filled$` `$arrow.t.bar$` `$arrow.t.quad$` `$arrow.t.dashed$` `$arrow.t.double$` `$arrow.t.stroked$` `$arrow.t.curve$`**Arrows Bottom** `$arrow.b$` `$arrow.b.triple$` `$arrow.b.filled$` `$arrow.b.bar$` `$arrow.b.quad$` `$arrow.b.dashed$` `$arrow.b.double$` `$arrow.b.stroked$` `$arrow.b.curve$`**Double Arrows Top Bottom** `$arrow.t.b$` `$arrow.t.b.filled$` `$arrow.t.b.double$` `$arrow.t.b.stroked$`**Arrows Diagonal Top Right** `$arrow.tr$` `$arrow.tr.filled$` `$arrow.tr.double$` `$arrow.tr.hook$` `$arrow.tr.stroked$`**Arrows Diagonal Bottom Right** `$arrow.br$` `$arrow.br.filled$` `$arrow.br.double$` `$arrow.br.hook$` `$arrow.br.stroked$`**Arrows Diagonal Bottom Left** `$arrow.bl$` `$arrow.bl.filled$` `$arrow.bl.double$` `$arrow.bl.hook$` `$arrow.bl.stroked$`**Arrows Diagonal Top Left** `$arrow.tl$` `$arrow.tl.filled$` `$arrow.tl.double$` `$arrow.tl.hook$` `$arrow.tl.stroked$`**Double Arrows Diagonal** `$arrow.tl.br$` `$arrow.tr.bl$`**Other Arrows** `$arrow.cw$` `$arrow.ccw.half$` `$arrow.cw.half$` `$arrow.ccw$`**7.2.9 Angles****Symbol Raw** `$angle$` `$angle.acute$` `$angle.l$` `$angle.r.double$` `$angle.right.arc$` `$angle.spheric$`**Symbol Raw** `$angle.rev$` `$angle.arc$` `$angle.r$` `$angle.right$` `$angle.right.dot$` `$angle.spheric.rev$`**Symbol Raw** `$angle.acute$` `$angle.arc.rev$` `$angle.l.double$` `$angle.right.rev$` `$angle.right.sq$` `$angle.spheric.top$`

7.2.10 Cool Symbols

Symbol	Raw	Symbol	Raw	Symbol	Raw
@	<code>\$at\$</code>	%	<code>\$co\$</code>	©	<code>\$copyright\$</code>
©	<code>\$copyright.sound\$</code>	°C	<code>\$degree.c\$</code>	€	<code>\$euro\$</code>
\$	<code>\$dollar\$</code>	£	<code>\$pound\$</code>	₩	<code>\$won\$</code>
¥	<code>\$yen\$</code>	฿	<code>\$bitcoin\$</code>	°F	<code>\$degree.f\$</code>
!	<code>\$excl\$</code>	¡	<code>\$excl.inv\$</code>	!!	<code>\$excl.double\$</code>
!?	<code>\$excl.quest\$</code>	↯	<code>\$arrow.zigzag\$</code>	⊗	<code>\$ast.circle\$</code>
⚙	<code>\$ast.triple\$</code>	χ	<code>\$chi\$</code>	🌸	<code>\$floral\$</code>
⚔	<code>\$maltese\$</code>	¶	<code>\$pilcrow\$</code>	ℏ	<code>\$planck\$</code>
♣	<code>\$suit.club\$</code>	◆	<code>\$suit.diamond\$</code>	♥	<code>\$suit.heart\$</code>
♠	<code>\$suit.spade\$</code>	△	<code>\$triangle.stroked.nested2\$</code>		

7.2.11 Style

Symbol	Raw	Symbol	Raw
$ABC123$	<code>\$sans(A B C 1 2 3)\$</code>	$\mathfrak{ABC}123$	<code>\$frak(A B C 1 2 3)\$</code>
$ABC123$	<code>\$mono(A B C 1 2 3)\$</code>	$\mathbb{ABC}123$	<code>\$bb(A B C 1 2 3)\$</code>
$\mathcal{ABC}123$	<code>\$cal(A B C 1 2 3)\$</code>		

Symbol	Raw
$\sum_{i \in \mathbb{N}} 1 + i$	<pre>#show math.equation: set text(font: "Fira Math") \$sum_(i in NN) 1 + i\$,</pre>

8 | Emoji Symbols

This is an incomplete list for all emoji goto [here](#)

If the emoji module is imported the `#emoji` can be removed

```
#import emoji: *
```

Sym

Raw

Sym

Raw

`#emoji.face`

```
#bibliography("../03-tail/bibliography.bib", style:"apa")  
#bibliography("../03-tail/bibliography.bib", style:"chicago-author-date")  
#bibliography("../03-tail/bibliography.bib", style:"chicago-notes")  
#bibliography("../03-tail/bibliography.bib", style:"ieee")  
#bibliography("../03-tail/bibliography.bib", style:"mla")
```

Bibliography

- [1] P. Fettke, “State-of-the-Art des State-of-the-Art,” *Wirtschaftsinformatik*, pp. 257–266, 2006, doi: 10.1007/s11576-006-0057-3.